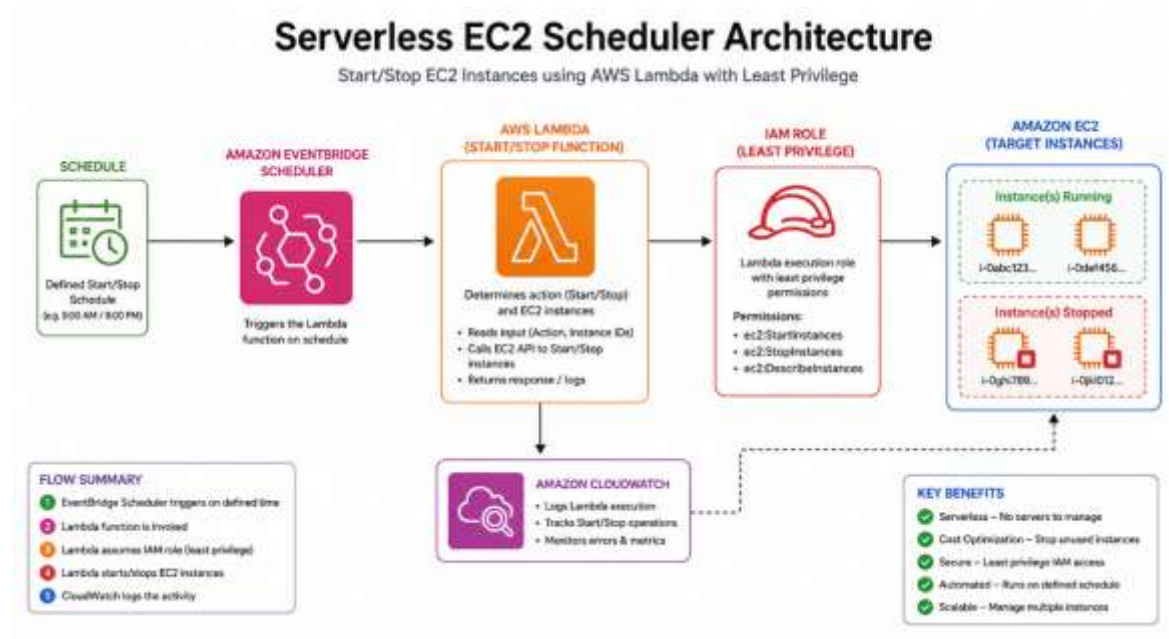
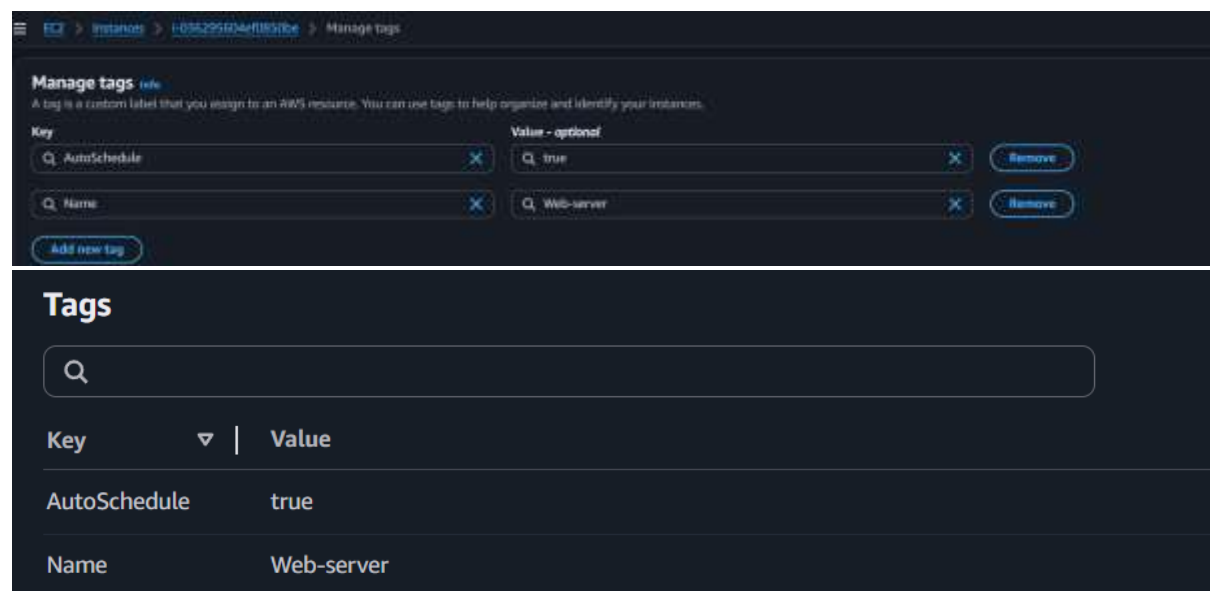


Serverless EC2 Scheduler (Start/Stop Instances using Lambda with Least Privilege)



Step 1: Tag Your EC2 Instances

- Add tag to instances:
- Key = AutoSchedule
- Value = true
- This avoids hardcoding instance IDs



Step 2: Create IAM Role (LEAST PRIVILEGE)

Lambda > Functions > Create function

Create function Info

Choose one of the following options to create your function.

☒ Author from scratch
Start with a simple Hello World example.

☐ Use a blueprint
Build a Lambda application from sample code and configuration presets for common use cases.

☐ Container image
Select a container image to deploy for your function.

Basic information

Function name
Enter a name that describes the purpose of your function.

EC2-start-stop

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime Info
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.12

EC2-start-stop-role-xxxzyzchi Info

Summary

Creation date
May 20, 2026, 18:32 (UTC+05:30)

Last activity
-

ARN
arn:aws:iam::749794722599:role/service-role/EC2-start-stop-role-xxxzyzchi

Maximum session duration
1 hour

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Trusted entities

Entities that can assume this role under specified conditions.

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Principal": {
7         "Service": "lambda.amazonaws.com"
8       },
9       "Action": "sts:AssumeRole"
10    }
11  ]
12 }
```

Edit trust policy

Permissions

Trust relationships

Tags

Last Accessed

Revoke sessions

Permissions policies (1) Info

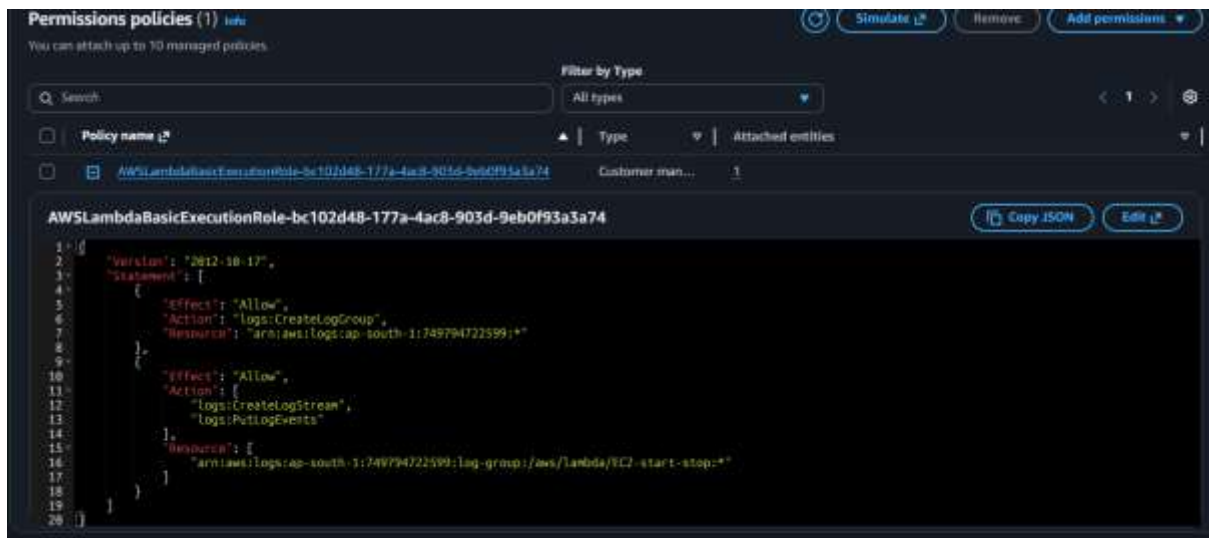
You can attach up to 10 managed policies.

Q Search

Filter by Type
All types

< 1 > ⚙

☐	Policy name Info	Type	Attached entities
☐	AWSLambdaBasicEct...	Customer man...	1

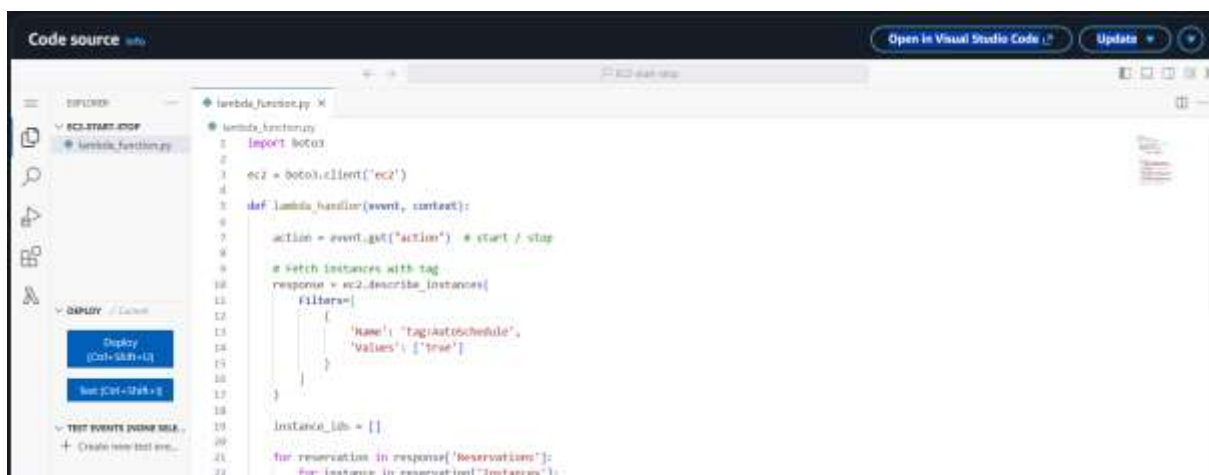


Why this is powerful:

Lambda can:

- ONLY start/stop tagged instances
- Cannot touch anything else

Step 3: Lambda Code (Production Ready)



```
import boto3
```

```
ec2 = boto3.client('ec2')
```

```
def lambda_handler(event, context):
```

```
    action = event.get("action") # start / stop
```

```
    # Fetch instances with tag
```

```
    response = ec2.describe_instances(
```

```
        Filters=[
```

```
            {
```

```

        'Name': 'tag:AutoSchedule',
        'Values': ['true']
    }
]
)

instance_ids = []

for reservation in response['Reservations']:
    for instance in reservation['Instances']:
        instance_ids.append(instance['InstanceId'])

if not instance_ids:
    return {"message": "No instances found"}

if action == "start":
    ec2.start_instances(InstanceIds=instance_ids)
return {"message": f"Started {instance_ids}"}

elif action == "stop":
    ec2.stop_instances(InstanceIds=instance_ids)
    return {"message": f"Stopped {instance_ids}"}

else:
    return {"error": "Invalid action"}

```

Step 4: Test Events

Start

```

{
    "action": "start"
}

```

Test event new

To invoke your function without saving an event, modify the event, then choose Test. Lambda uses the modified event to invoke your function, but does not overwrite the original event until you choose Save.

Test event action

☐ Create new event ☒ Edit saved event

Invocation type

☐ Synchronous
Invokes the Lambda function and blocks until receiving the function's response, with a maximum timeout of 15 minutes. Returns function output or error details directly to the calling application.

☒ Asynchronous
Executes the Lambda function for invocation and returns immediately with a request ID. Function proceeds independently, with results optionally sent to a configured destination like SQS, SNS, or EventBridge.

Event name

start-EC2

Event JSON

```

{
  "action": "start"
}

```

Format JSON

Buttons: Delete, CloudWatch Logs Live Tail, Save, Test

Stop

```

{
    "action": "stop"
}

```

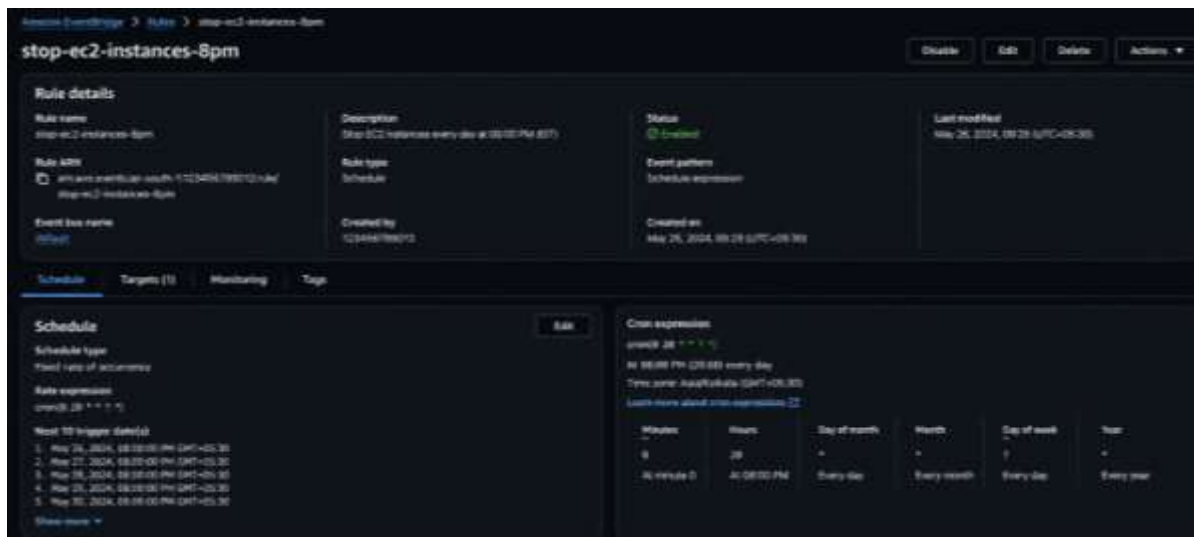
Step 5: EventBridge Scheduler

Create 2 Rules

1. Stop Rule

Cron: 0 20 * * ? *

8 PM → stop instances



2. Start Rule

Cron: 0 8 * * ? *

8 AM → start instances

EC2 INSTANCES – RUNNING (After Start by Scheduler - 8:00 AM)

New EC2 Experience

EC2 Dashboard

EC2 Global View

Events

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Instances (2)

Find instance by attribute or tag name

Refresh

Instance state

Actions

Launch instances

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP v4 (IPv4)	Public IP v6
☐	Web Server	i-6ab125cd456ef7890	Running	t3.micro	1/2 checks passed	View alarms	ap-south-1a	ec2-15-206-xxx-xxx.ap-s...	15.206.xxx.x
☐	App Server	i-6ab125cd456ef7890	Running	t3.micro	1/2 checks passed	View alarms	ap-south-1a	ec2-15-206-xxx-xxx.ap-s...	15.206.xxx.x

Instance: i-6ab125cd456ef7890 (Web-Server)

Details

Status and alarms

Tags

Monitoring

Security

Networking

Storage

Logs

Instance monitoring

Instance ID: i-6ab125cd456ef7890 (Web-Server)

Public IP v4 address: 15.206.xxx.xxx | [View address](#)

Private IP v4 address: 172.31.12.48

Public IP v6: ec2-15-206-xxx-xxx.ap-south-1.compute.amazonaws.com | [View address](#)

STATUS: RUNNING

Instances started successfully by EventBridge Scheduler at 08:00 AM (IST)